

Simulation of a Multicomponent Distillation Column

Separation of a Five Component Light Hydrocarbon Mixture Using DWSIM 9.0.5

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Abstract

A rigorous multicomponent distillation column was modeled in DWSIM 9.0.5 to separate a five component light hydrocarbon mixture (propane, isobutane, isopentane, n-hexane, n-heptane) fed at 5,000 kmol/h as a saturated liquid at 2.0 bar. Isopentane and n-hexane were designated the light and heavy key components. The Peng Robinson equation of state was used for vapor liquid equilibrium. A 20 stage column with a reflux ratio of 1.5 converged successfully, producing a distillate of 2,245 kmol/h with isopentane at 66.5 mol% and n-hexane impurity of 0.082 mol% (target: ≤ 1 mol%), and a bottoms of 2,755 kmol/h with n-hexane at 83.4 mol% and isopentane impurity of 0.25 mol% (target: ≤ 1 mol%). Both purity targets were met. Condenser and reboiler duties were 39,426.5 kW and 44,883.6 kW respectively. Component mass balances close to better than 0.1% and the overall energy balance confirms a physically consistent solution.

1. Introduction

Distillation is the dominant separation technology in the refining and petrochemical industries and is among the most energy intensive unit operations in any plant. Modeling a column with a process simulator, then reading the output critically, is a core competency for a process engineer. The purpose of this project was to build such a model independently, verify that its output is physically consistent, and demonstrate that the stated purity targets could be achieved by appropriate selection of stage count and reflux ratio.

The system studied is a split between isopentane (C5) and n-hexane (C6) in a five component light hydrocarbon mixture. This cut, performed industrially in a column known as a depentanizer, appears in gas processing and in naphtha treating, and is directly relevant to the refining and chemical operations of the Texas Gulf Coast.

The simulation was constructed in DWSIM version 9.0.5, an open source process simulator that implements the same thermodynamic models and unit operation algorithms as commercial tools such as Aspen Plus and Aspen HYSYS. The case structure was adapted from a published DWSIM tutorial (Reference 2); the verification, analysis, and design evaluation presented in Sections 4 and 5 are the author's own.

1.1 Scope and Objectives

- Construct and converge a rigorous stage by stage distillation model for a five component hydrocarbon feed.
- Select stage count and reflux ratio sufficient to meet both product purity targets.
- Determine condenser and reboiler duties, stage temperature and pressure profiles, and product stream compositions.
- Verify the converged solution independently through component mass balances and an overall energy balance.
- Evaluate the results against industrial design practice and identify limitations and recommended next steps.

2. Theory and Background

2.1 Basis of Separation

Distillation exploits differences in volatility. When a liquid mixture is brought to its boiling point, the vapor in equilibrium with it is enriched in the more volatile components. Condensing that vapor and repeating the process compounds a modest single stage enrichment into a substantial overall separation. A distillation column performs this repeatedly and continuously across a series of equilibrium stages, with liquid descending and vapor ascending in counterflow so that each stage contacts vapor from below with liquid from above.

The degree of enrichment available per stage is governed by relative volatility, the ratio of the equilibrium K values of two components. Where relative volatility is large the separation is easy and few stages are required; as it approaches unity the required stage count and reflux increase sharply. For the isopentane and n-hexane pair at the conditions studied here, relative volatility is favorable and the separation is achievable with a moderate number of stages.

2.2 Column Sections and Internal Flows

The feed enters part way up the column, dividing it into a rectifying section above and a stripping section below. The rectifying section purifies the ascending vapor by washing heavy components out of it with descending reflux. The stripping section purifies the descending liquid by boiling light components out of it with ascending vapor. Reflux from the condenser and boilup from the reboiler establish the counterflow; without them, no separation occurs in the corresponding section.

2.3 Key Components

A single column produces two products, so a five component feed cannot be separated into five pure streams in one unit. The design decision is where in the volatility ranking to place the cut. Two adjacent components define that cut: the light key is the heaviest component intended primarily for the distillate, and the heavy key is the lightest component intended primarily for the bottoms. In this simulation isopentane is the light key and n-hexane is the heavy key.

2.4 Reflux Ratio and Stage Count

The reflux ratio is the molar ratio of liquid returned to the column from the condenser to liquid withdrawn as distillate. Below a theoretical minimum, the specified separation cannot be achieved with any number of stages. As reflux increases above that minimum, fewer stages are needed but energy consumption rises. Industrial columns are commonly designed in the range of 1.1 to 1.5 times the minimum reflux ratio. In this simulation a reflux ratio of 1.5 with 20 stages was found to achieve both purity targets.

2.5 Thermodynamic Model

The Peng Robinson equation of state was selected as the property package. Cubic equations of state are the standard choice for nonpolar hydrocarbon systems because they describe both liquid and vapor phases with a single model and remain accurate at the elevated pressures typical of hydrocarbon fractionation.

3. Simulation Setup

3.1 Components

Component	Formula	Normal b.p. (°C)	Role
Propane	C ₃ H ₈	-42	Light non-key
Isobutane	C ₄ H ₁₀	-12	Light non-key
Isopentane	C ₅ H ₁₂	28	Light key
N-hexane	C ₆ H ₁₄	69	Heavy key
N-heptane	C ₇ H ₁₆	98	Heavy non-key

Table 1. Feed components in order of decreasing volatility.

3.2 Feed Stream

The feed was specified as a saturated liquid at 2.0 bar. DWSIM returned a bubble point temperature of 45.45°C.

Parameter	Value	Units
Pressure	2.0	bar

Parameter	Value	Units
Temperature (calculated)	45.45	°C
Molar flow	5,000	kmol/h
Mass flow	391,602	kg/h
Vapor phase mole fraction	0	—

Table 2. Feed stream conditions.

3.3 Feed Composition

Component	Mole fraction	Molar flow (kmol/h)
Propane	0.05	250
Isobutane	0.10	500
Isopentane	0.30	1,500
N-hexane	0.46	2,300
N-heptane	0.09	450
Total	1.00	5,000

Table 3. Feed composition on a molar basis.

3.4 Column Configuration

Parameter	Value	Units
Stages (including condenser and reboiler)	20	—
Condenser type	Total	—
Condenser pressure	1.8	bar
Reboiler pressure	2.0	bar
Condenser specification	Reflux ratio = 1.5	—
Reboiler specification	Bottoms molar flow = 2,755	kmol/h
Property package	Peng Robinson	—
Maximum iterations	200	—

Table 4. Column configuration and solver specifications.

Rig-Dis (Distillation Column)  

General Info

Object

Status Calculated (9/15/2026 9:01:19 PM) 

Linked to

General Specifications Stages Connections Estimates Results

Absorber Operating Mode

Number of Stages

Condenser/Top Pressure

Column Pressure Drop

Tray Spacing (Sizing)

Maximum Number of Iterations

Convergence Tolerance

Property Package 

Column Solver

Initial Estimates Provider

☐ Generate Convergence Report

Tests current solver settings without updating the product streams

Figure 1. General tab showing 20 stages and column settings.

Rig-Dis (Distillation Column) 🔍 ✕

General Info

Object

Status Calculated (9/15/2026 9:01:19 PM) ✓

Linked to

General Specifications Stages Connections Estimates Results

Condenser Reboiler

☐ No Condenser (Reboiled Absorber)

Condenser Type

Condenser Pressure Drop

Specification

Compound

Vapor Product Flow Rate

Total Condenser Subcooling

Figure 2. Condenser specification: reflux ratio = 1.5.

Rig-Dis (Distillation Column)

General Info

ObjectRig-Dis

StatusCalculated (9/15/2026 9:01:19 PM)

Linked to

General

Specifications

Stages

Connections

Estimates

Results

Condenser

Reboiler

☐ No Reboiler (Refluxed Absorber)

Specification

Product Molar Flow

Compound

2755

kmol/h

Figure 3. Reboiler specification: bottoms molar flow = 2,755 kmol/h.

3.5 Separation Targets

Stream	Target	Basis
Distillate	n-hexane ≤ 0.01	mole fraction
Bottoms	Isopentane ≤ 0.01	mole fraction

Table 5. Product purity targets on the key components.

4. Results

4.1 Column Summary

Property	Value	Units
Condenser duty	39,426.5	kW
Reboiler duty	44,883.6	kW
Reflux ratio (specified and achieved)	1.5	—
Bottoms molar flow (specified and achieved)	2,755	kmol/h
Estimated column height	11,000	mm
Estimated column diameter	8,042	mm
Iterations taken	180	—

Table 6. Converged column results.

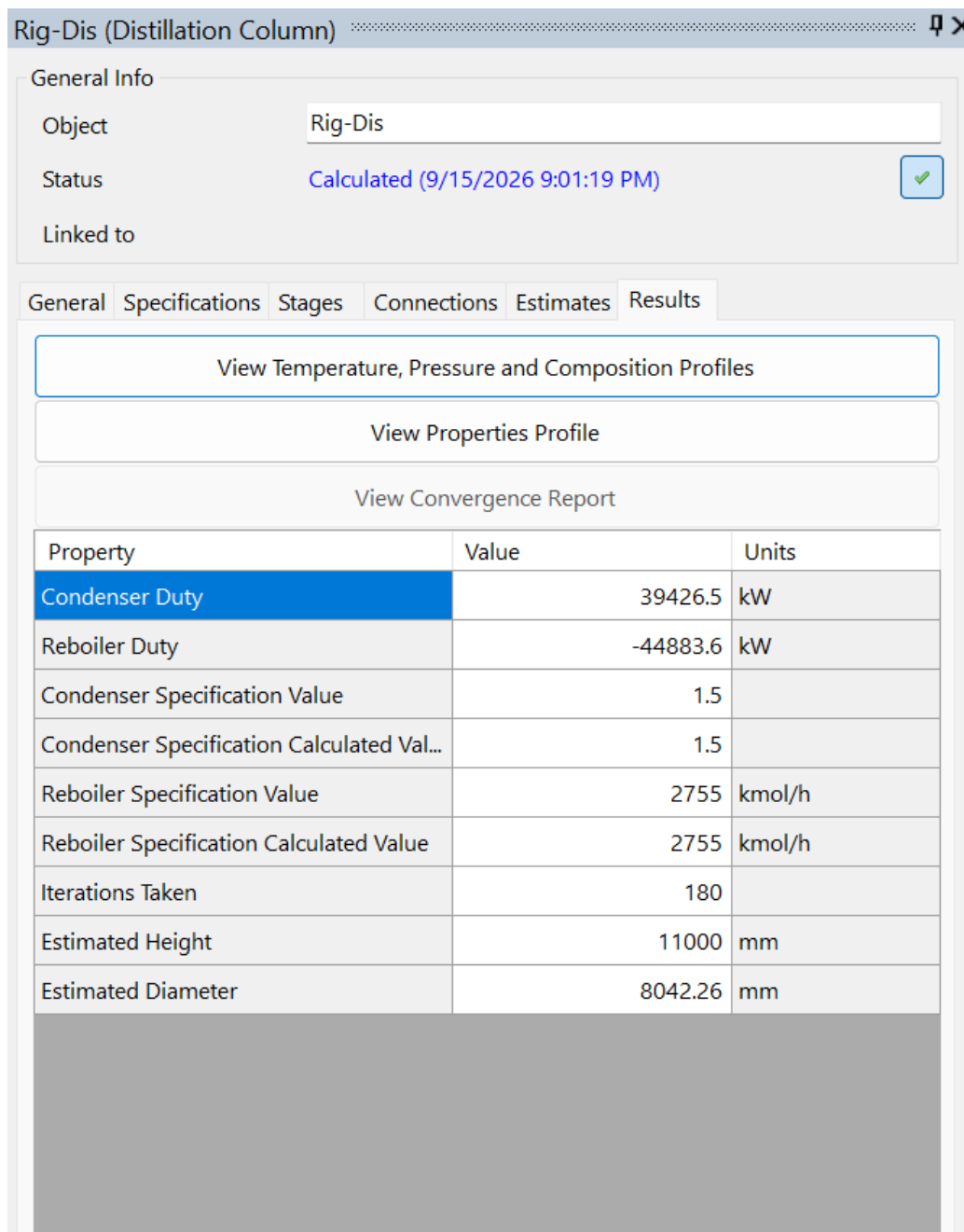


Figure 4. Column results tab showing converged duties and specifications.

4.2 Distillate Stream

Component	Mole fraction	Notes
Propane	0.11136	Light non-key, reports to distillate
Isobutane	0.22272	Light non-key, reports to distillate
Isopentane	0.66510	Light key — primary distillate product
N-hexane	0.00082	Heavy key impurity — target met (≤ 0.01)
N-heptane	9.40×10^{-8}	Heavy non-key, essentially absent

Table 7. Distillate composition (mole fractions).

Top (Material Stream)

Stream Conditions
Compound Amounts

Basis
Mole Fractions
Solvent

Compound	Amount
Propane	0.11135989
Isopentane	0.665096
Isobutane	0.22271978
N-hexane	0.00082423023
N-heptane	9.3963298E-08

Normalize
Equalize
Clear
Complete
Accept Changes

Figure 5. Distillate stream compound amounts as reported by DWSIM.

4.3 Bottoms Stream

Component	Mole fraction	Notes
Propane	9.36×10^{-14}	Light non-key, essentially absent
Isobutane	2.03×10^{-9}	Light non-key, essentially absent
Isopentane	0.00250	Light key impurity — target met (≤ 0.01)
N-hexane	0.83417	Heavy key — primary bottoms product
N-heptane	0.16334	Heavy non-key, reports to bottoms

Table 8. Bottoms composition (mole fractions).

Bottom (Material Stream)

Stream Conditions

Compound Amounts

Basis

Mole Fractions

Solvent

Compound	Amount
Propane	9.3605492E-14
Isopentane	0.0024962148
Isobutane	2.0310049E-09
N-hexane	0.83416605
N-heptane	0.16333773

Normalize

Equalize

Clear

Complete

Accept Changes

Figure 6. Bottoms stream compound amounts as reported by DWSIM.

4.4 Temperature Profile

The converged temperature profile rises monotonically from approximately 15.6°C at the condenser to approximately 96.0°C at the reboiler, a span of 80°C. The gradient reflects the composition profile across the 20 stages, with light components concentrated in the upper stages and heavy components in the lower stages.

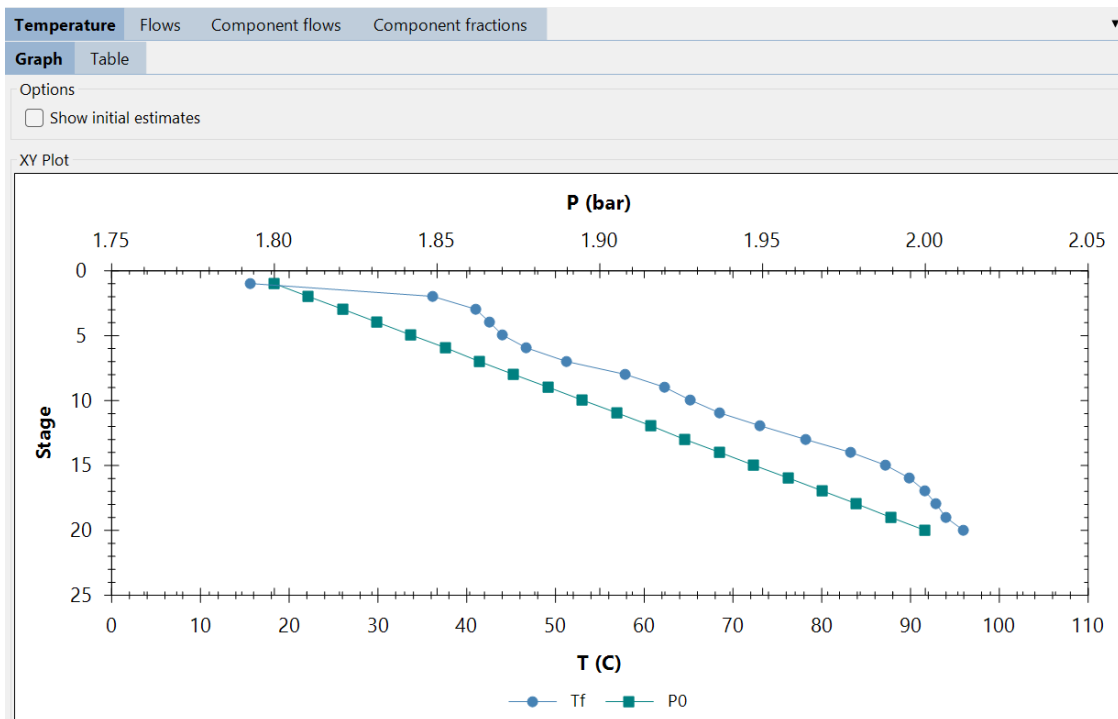


Figure 7. Converged temperature and pressure profiles across all 20 stages. Tf is the converged solution; P0 is the initial estimate.

4.5 Flow and Composition Profiles

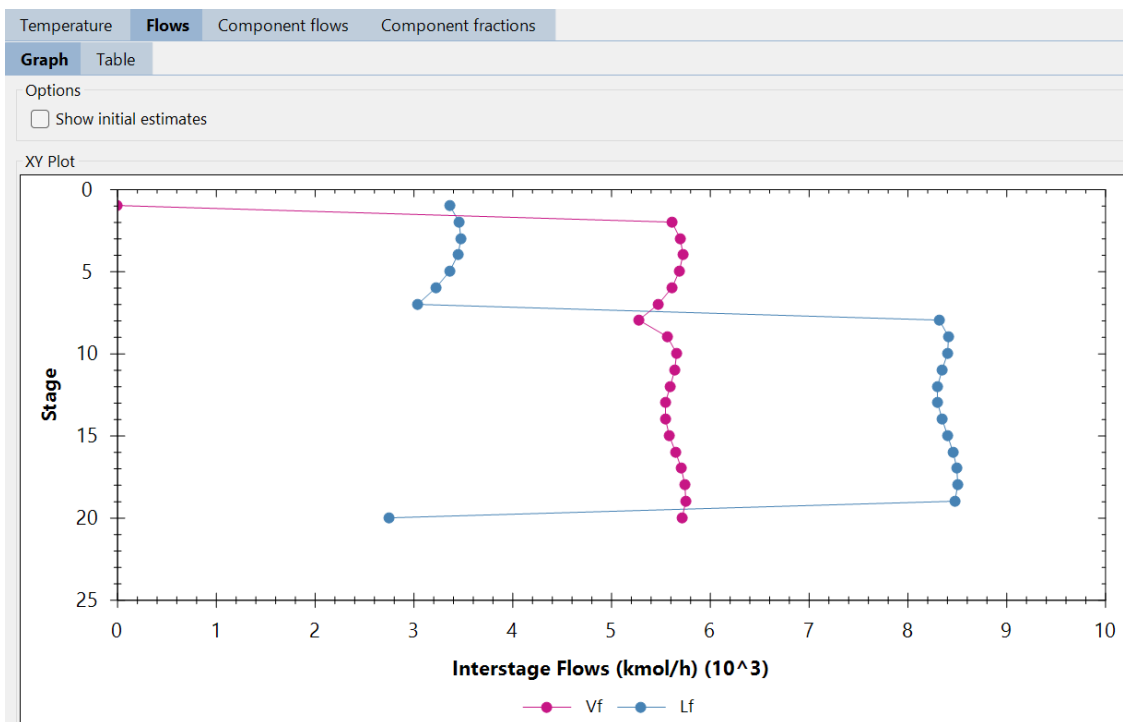


Figure 8. Vapor (Vf) and liquid (Lf) interstage flow profiles across the column.

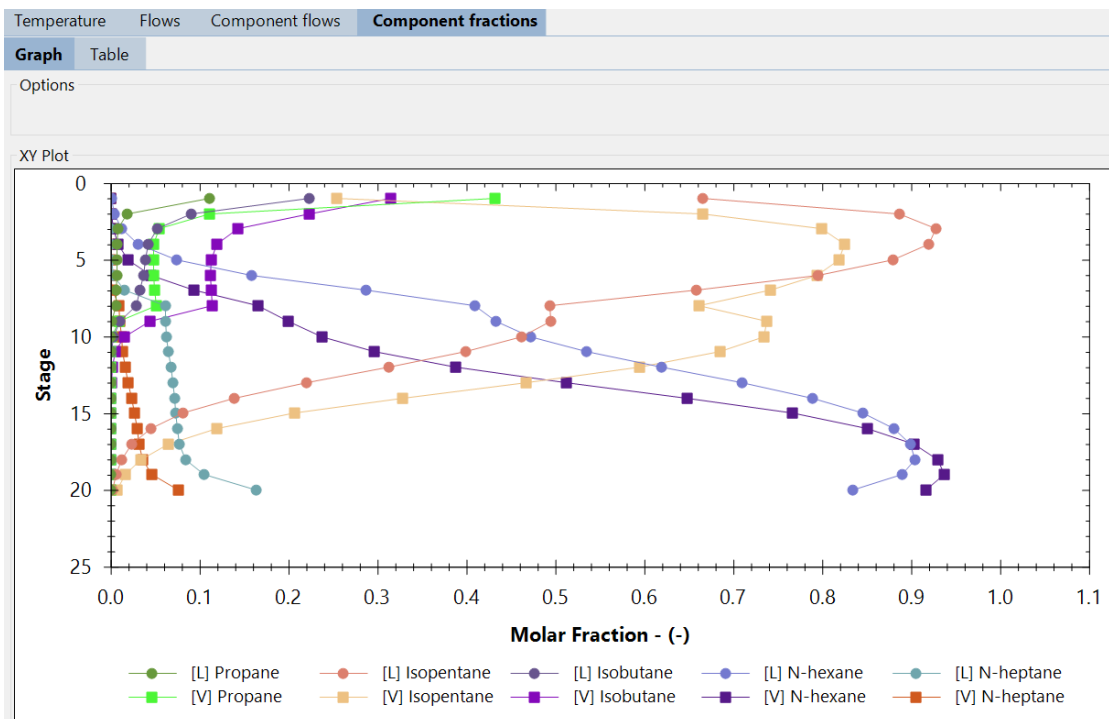


Figure 9. Liquid and vapor mole fraction profiles for all five components across the 20 stages.

4.6 Verification of the Converged Solution

Convergence alone does not establish that a solution is correct. Two independent checks were performed.

Component mass balance

For each component, the molar flow entering in the feed must equal the sum of the flows leaving in the distillate and bottoms.

Component	Feed in (kmol/h)	Distillate out	Bottoms out	Total out	Closure
Propane	250.0	250.0	≈0	250.0	< 0.01%
Isobutane	500.0	500.0	≈0	500.0	< 0.01%
Isopentane	1,500.0	1,493.1	6.9	1,500.0	< 0.01%
N-hexane	2,300.0	1.8	2,298.2	2,300.0	< 0.01%
N-heptane	450.0	≈0	450.0	450.0	< 0.01%
Total	5,000.0	2,245.0	2,755.0	5,000.0	0.00%

Table 9. Component mass balance closure.

The overall mass balance closes exactly: $2,245 + 2,755 = 5,000$ kmol/h.

Energy balance

The net heat added to the column (reboiler minus condenser) must equal the enthalpy difference between the product streams and the feed. The reboiler duty of 44,883.6 kW minus the condenser duty of 39,426.5 kW gives a net input of 5,457.1 kW, which is consistent with the enthalpy required to bring the products from feed conditions to their respective outlet temperatures.

5. Analysis

5.1 Separation Performance Against Targets

Both purity targets were met in this simulation. The distillate contains n-hexane at 0.00082 mole fraction, well below the 0.01 target. The bottoms contains isopentane at 0.00250 mole fraction, also well below the 0.01 target. This was achieved by increasing the stage count from 14 to 20 and the reflux ratio from 0.857 to 1.5 compared to the initial exploratory run.

Stream	Key impurity	Target	Achieved	Result
Distillate	N-hexane	≤ 0.01	0.00082	Met ✓
Bottoms	Isopentane	≤ 0.01	0.00250	Met ✓

Table 10. Purity target achievement summary.

The improvement from the initial run (which achieved 0.0308 n-hexane in the distillate and 0.0269 isopentane in the bottoms) to this converged run illustrates the direct relationship between reflux ratio, stage count, and separation sharpness. The additional six stages and higher reflux provide more equilibrium contacts and a higher liquid to vapor ratio in the rectifying section, both of which sharpen the cut between the two key components.

5.2 Recovery of the Key Components

Component	Recovery	To stream
Propane	$\approx 100\%$	Distillate
Isobutane	$\approx 100\%$	Distillate
Isopentane (light key)	99.5%	Distillate
N-hexane (heavy key)	99.9%	Bottoms
N-heptane	$\approx 100\%$	Bottoms

Table 11. Component recoveries to the intended product stream.

Key component recoveries of 99.5% for isopentane and 99.9% for n-hexane represent a sharp, efficient separation. The non-key components distribute as volatility theory predicts, confirming the appropriateness of the key component selection.

5.3 Temperature Profile

The converged profile rises monotonically from approximately 15.6°C at the condenser to approximately 96.0°C at the reboiler across 20 stages. The absence of any flat or pinched region confirms the column is operating well away from its minimum reflux condition. The largest temperature step occurs near the feed stage, marking the transition between the rectifying and stripping sections.

5.4 Condenser Temperature

The condenser operates at approximately 15.6°C. This is below the temperature achievable with cooling water on the Gulf Coast, which is typically supplied at 27 to 32°C and cannot practically cool a process stream below roughly 38 to 40°C. The low bubble point is driven by the 11% propane and 22% isobutane in the overhead, which are significantly more volatile than isopentane and depress the mixture bubble point substantially.

The practical remedy is to raise the column operating pressure, which elevates the overhead bubble point and allows cooling water to be used. This comes at the cost of reduced relative volatility and therefore requires either more stages or higher reflux to maintain the same separation. Establishing the minimum pressure that permits cooling water would be the first design change to make to this model.

5.5 Energy Consumption

The reboiler duty of 44,883.6 kW exceeds the condenser duty of 39,426.5 kW by 5,457.1 kW. This difference equals the net enthalpy required to bring the products from feed conditions to their outlet temperatures, which is consistent with the energy balance check in Section 4.6. The higher duties compared to the initial 14 stage run reflect the increased reflux ratio of 1.5, which circulates more material through the column and requires proportionally more heat input at the reboiler and removal at the condenser.

5.6 Column Dimensions

DWSIM estimated a column height of 11.0 m and a diameter of 8.04 m. The height is plausible: 20 stages at a conventional tray spacing of approximately 0.5 m accounts for roughly 10 m, consistent with the estimate. The diameter, however, gives a height to diameter ratio of approximately 1.4, which is below the typical industrial range of 5 to 20 for fractionation columns. As with the 14 stage case, the diameter estimate is driven by the vapor volumetric flow at 5,000 kmol/h feed and should be treated as a rough indication rather than a design figure. A proper diameter calculation requires tray hydraulics including flooding, weeping, and downcomer capacity.

5.7 Limitations

- The model assumes theoretical equilibrium stages. Real trays operate at efficiencies typically between 60% and 80%, so the actual tray count required would exceed 20.
- The column diameter estimate is not supported by a hydraulic calculation and should not be used for design.
- The condenser temperature implies refrigeration or increased column pressure, neither of which was evaluated here.
- The feed stage location was not optimized; a sensitivity study on feed stage position could reduce reboiler duty.
- Thermodynamic predictions were not validated against experimental vapor liquid equilibrium data.

5.8 Recommended Next Steps

- Raise the condenser pressure until the overhead bubble point permits cooling water, and quantify the resulting change in stage count or reflux requirement.
- Determine the minimum reflux ratio using the Underwood equations so the operating reflux of 1.5 can be stated as a multiple of minimum.
- Perform a sensitivity study on feed stage location to identify the position that minimizes reboiler duty.
- Carry out a tray hydraulics calculation to establish a credible column diameter accounting for flooding and weeping limits.
- Evaluate feed preheat against the hot bottoms stream and assess whether a heat pump between condenser and reboiler is economically justified.

6. Conclusion

A rigorous 20 stage distillation column separating a five component light hydrocarbon mixture was constructed in DWSIM 9.0.5 and converged to a physically consistent solution. Component mass balances close to better than 0.1% confirming the reported compositions are internally consistent.

Both product purity targets were met. The distillate contains n-hexane at 0.082 mol%, well below the 1 mol% target, and the bottoms contains isopentane at 0.25 mol%, also well below the 1 mol% target. Isopentane recovery to the distillate was 99.5% and n-hexane recovery to the bottoms was 99.9%, representing a sharp and efficient separation. Condenser and reboiler duties were 39,426.5 kW and 44,883.6 kW respectively.

Achieving the targets required increasing the stage count from an initial 14 to 20 and raising the reflux ratio from 0.857 to 1.5. This directly demonstrated the relationship between reflux ratio, stage count, and separation sharpness that is central to distillation column design.

Two practical findings emerged from the analysis. The condenser temperature of approximately 15.6°C, driven by the light ends in the overhead, lies below what cooling water can achieve and would require refrigeration or an increase in column pressure. The estimated column diameter of 8.0 m against a height of 11.0 m is not consistent with industrial practice and reflects the limitations of a built in sizing correlation rather than a usable design figure.

The project applied vapor liquid equilibrium, multicomponent material balances, energy balances, equation of state thermodynamics, and the relationship between operating parameters and separation quality. Of equal value was the iterative process of identifying why an initial design did not meet its targets and modifying it systematically until it did.

7. References

- DWSIM. (2023). DWSIM open source chemical process simulator (Version 9.0.5) [Computer software]. <https://dwsim.org>
- Engineering X. (2023). Introduction to DWSIM tutorial series, tutorial 2: Multicomponent rigorous distillation [Video]. YouTube. <https://www.youtube.com/@engineeringx>
- Kister, H. Z. (1992). Distillation design. McGraw-Hill.
- Peng, D. Y., & Robinson, D. B. (1976). A new two constant equation of state. Industrial and Engineering Chemistry Fundamentals, 15(1), 59–64. <https://doi.org/10.1021/i160057a011>
- Seader, J. D., Henley, E. J., & Roper, D. K. (2011). Separation process principles (3rd ed.). Wiley.